Exact Recurrences and Monotonicity for Prime Partitions

A note on OEIS A000607

Primary author (AI): ChatGPT (OpenAI)

Prepared November 13, 2026. AI-authored mathematical exposition; checked by internal derivation and finite computation, not independently peer reviewed.

Abstract

Let $a(n)$ count unordered partitions of n into prime parts, with $a(0) = 1$. We prove its Euler-product generating function, an exact divisor-sum recurrence, a multiplicative-partition correspondence, and strict two-step monotonicity for $n \geq 3$. These are classical or elementary consequences of the definition, not a resolution of an open conjecture.

1 The OEIS definition and an exact recurrence

The sequence A000607 counts partitions into primes with unlimited repetitions and with order ignored. For example, the five partitions of 10 are

$$2+2+2+2+2, \quad 2+2+3+3, \quad 2+3+5, \quad 3+7, \quad 5+5.$$

The empty partition is counted at $n = 0$.

Theorem 1 (Prime-partition product and recurrence)**.** *In the ring of formal power series* $\mathbb{Z}[[z]]$*,*

$$F(z) := \sum_{n\geq 0} a(n) z^n = \prod_{p \text{ prime}} (1 - z^p)^{-1}. \tag{1}$$

If $b(k) := \sum_{\substack{p \text{ prime} \\ p|k}} p$ *(distinct prime divisors), then*

$$a(0) = 1, \qquad a(n) = \frac{1}{n} \sum_{k=1}^{n} b(k) a(n-k) \quad (n \geq 1). \tag{2}$$

Proof. Each factor in (1) expands as $1 + z^p + z^{2p} + \cdots$ and records the multiplicity of part p. Every coefficient depends on only finitely many factors, so the infinite product is well defined formally. Formal logarithmic differentiation is legitimate coefficientwise and gives

$$\frac{zF'(z)}{F(z)} = \sum_{p} \frac{pz^p}{1 - z^p} = \sum_{p} \sum_{r\geq 1} p z^{rp} = \sum_{k\geq 1} b(k) z^k.$$

Multiplying by $F(z)$ and comparing the coefficient of z^n gives $na(n) = \sum_{k=1}^{n} b(k) a(n-k)$, proving (2). □

2 A second interpretation and a monotonicity theorem

Proposition 1 (Prime-factor sum correspondence)**.** *Let* $\mathrm{sopfr}(m)$ *denote the sum of all prime factors of* m*, counted with multiplicity, and put* $\mathrm{sopfr}(1) = 0$*. Then*

$$a(n) = \#\{m \geq 1 : \mathrm{sopfr}(m) = n\}.$$

Proof. A multiset of prime parts with multiplicities e_p has sum $\sum_p pe_p = n$ and determines the positive integer $m = \prod_p p^{e_p}$. By the fundamental theorem of arithmetic this correspondence is bijective and its prime-factor sum is n. □

Theorem 2 (Strict two-step monotonicity)**.** *For every integer $n \geq 3$, one has $a(n+2) > a(n)$. The inequality is not strict for $n = 0$ or $n = 2$.*

Proof. Adding one part 2 to any partition of n injects its partitions into those of $n+2$. To show that it misses at least one partition for $n \geq 3$, it suffices to partition each integer $N \geq 5$ into odd primes. This is possible for $N = 5, 6, 7$ using 5, $3+3$, and 7, respectively. If $N \geq 8$, use an odd-prime partition of $N - 3 \geq 5$ and append 3. Induction produces a partition of N with no part 2, not in the image of the injection. Finally, $a(0) = a(2) = 1$ and $a(2) = a(4) = 1$. □

3 Verification, provenance, and scope

The recurrence yields

$$(a(0), \ldots, a(15)) = (1, 0, 1, 1, 1, 2, 2, 3, 3, 4, 5, 6, 7, 9, 10, 12),$$

matching the beginning of the OEIS entry. A separate exact-integer computation checked the recurrence through $n = 150$ and strict two-step monotonicity in that range. Such finite tests are checks, not substitutes for the proofs.

The generating product, arithmetic correspondence, and recurrence are already reflected in the OEIS literature; the argument here is expository. No assertion of priority or proof of an open prime-partition conjecture is made.

References

[1] OEIS Foundation, *A000607: Number of partitions of n into prime parts*, https://oeis.org/A000607 (accessed November 13, 2026).

[2] G. E. Andrews, *The Theory of Partitions*, Addison-Wesley, 1976 (general generating-function methodology).

[3] OEIS Foundation, *A001414: Integer logarithm: sum of prime factors of n with multiplicity*, https://oeis.org/A001414.